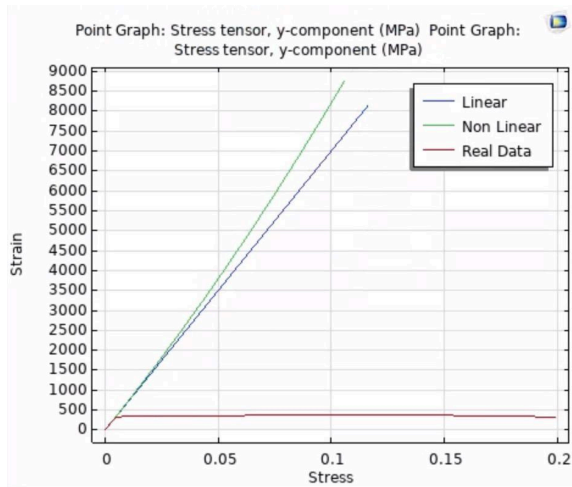


LeRoy Horton and Charlie Rahdon Project Assessment
ES120: Introduction to Mechanics of Solids

Assumption 1: Linear Approximation of Full Non-Linear Equations of Elasticity

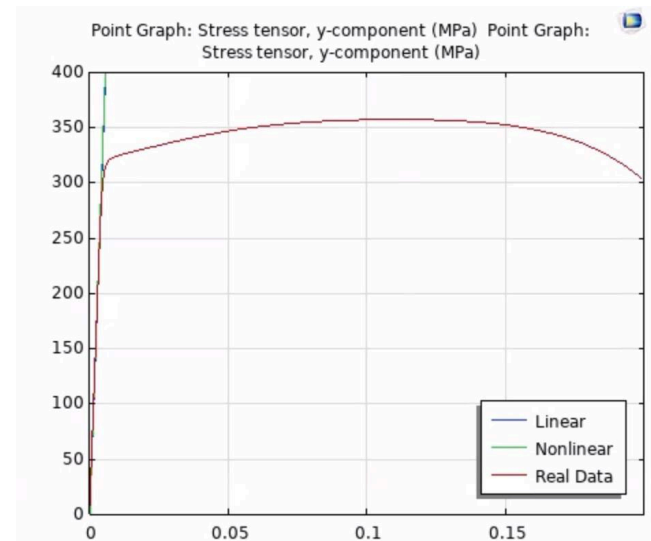
Overview: In order to analyze the difference between the linear approximation of the non-linear equations of elasticity and the non-linear equations of elasticity themselves, we used COSMOL to simulate the stress-strain relationship on a cylindrical shaft. A cylindrical shaft was the easiest shape to test this because the stress along the shaft should be equivalent.



Process: In the simulation, in order to evaluate the assumption, an axial force of 100kN was applied to an aluminium dogbone using COSMOL tools.

Afterwards, we measured the stress and strain along the simulated shaft. With that data we created a stress strain graph using the linear approximation to generate the graph, then we did the same but using the full non-linear equations, and finally we also added the results of the real stress strain curve from the aluminium dogbone in lab 1.

Conclusion: Looking at the data gathered, we concluded that the assumption using linear relationship, in the elastic region, of the stress strain curve used in engineering instead of the full non-linear equations proves reasonable within the portion of the graph pertinent to engineering. Although the two equations do differ shown in the image, when looking closely in comparison to the real data, in the elastic range which is the only range that we apply these equations to, the two different equations result in equal values. This allows us to conclude that while the linear and non linear models differ, this difference only occurs well out of the real life stress range that materials would experience, proving that linearizing the equations of elasticity is a valid approximation of the actual elasticity of a material.

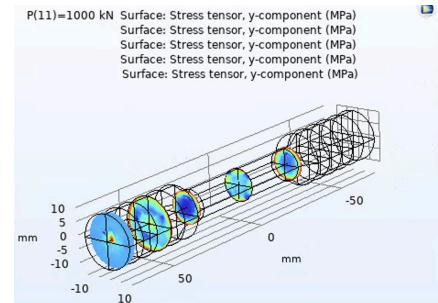


Assumption 2: Saint-Venant's Principle

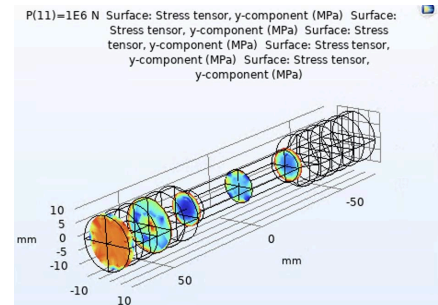
Overview: St Venant's Principle states that stress distributions resulting from statically equivalent loads become indistinguishable at locations sufficiently far from where the load is applied.

Process: In order to test this assumption, we modelled the same cylindrical dogbone sample in COMSOL and applied three boundary conditions: a point force, a distributed force applied across the front face and a force applied along the side. All of these boundary conditions had equivalent forces which were all applied axially in the y direction, and by taking five cross sections at different y planes in the dogbone ($y=-30\text{mm}$, $y=0\text{mm}$, $y=30\text{mm}$, $y=50\text{mm}$ and $y=70\text{mm}$), we were able to analyse how the stress distribution varies at different distances from the force.

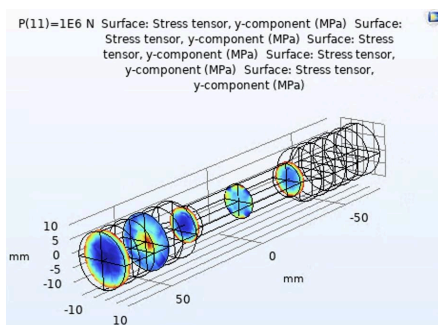
Conclusion: Near the load application ($y=70\text{mm}$), the stress distributions differ significantly between loading cases, with the point and face loads producing similar magnitudes while the side load shows a much lower value, indicating strong sensitivity to how the load is applied. Despite this, we found that when moving away from the load ($y=50\text{mm}$ and $y=30\text{mm}$), the stress values rapidly converge, and by the center of the specimen ($y=0\text{mm}$) and beyond ($y=-30\text{mm}$), all three loading cases produce nearly identical stresses. This convergence demonstrates that the influence of the boundary conditions diminishes with distance, and the stress state becomes dependent only on the resultant force, which provides clear support for St. Venant's principle.



Point force



Face force



Side force

	Average Cross-sectional Stress						
Type of Force	Location Along the Dogbone (mm)						Stress (MPa)
		-30	0	30	50	70	
	Point	7611	8150	7613	3182	3152	
	Face	7611 MPa	8150	7613	3183	3183	
	Sides	7611	8150	7613	3183	1140	